Measuring Strategic Hedging in Latin America: A Multilayer Network Approach

PhD research proposal

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Research Problem

The measurement of alignment strategies among weaker states has advanced significantly since Sims and Lee's (2025) Hedging Index for Latin America, which blends bilateral trade shares with arms procurement data using SIPRI's Trend Indicator Value series. While methodologically innovative, the index conflates three analytically distinct phenomena: structural legacy dependency, reactive adaptation to external actor behaviour, and genuine strategic hedging. The Nicaragua case illustrates this most starkly: a government openly hostile to Washington scores 0.946 on the US-alignment scale, an almost identical score to Honduras, which hosts a permanent US military base. This result reflects not strategic alignment but the invisibility of Russia as Nicaragua's actual arms supplier within a strictly bilateral US-China measurement framework. More broadly, the index treats trade flows and arms sales as signals of political alignment when they are better understood as products of market structure and supplier dominance.

This research proposes to construct a superior alignment measurement framework for Latin America that addresses these limitations, drawing on a richer and more theoretically grounded set of indicators, and applying multilayer network analysis to map the evolving structure of great power competition in the hemisphere.

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The intensification of great power rivalry between the United States, China, and Russia has placed weaker states under mounting pressure to align with one pole or another, making the study of alignment strategies one of the most pressing questions in contemporary international relations. Scholars have long examined how secondary states navigate this pressure through three primary strategies: balancing, which involves forming coalitions to counter the strongest power (Walt, 1987; Waltz, 1979) bandwagoning, which involves aligning with a dominant or threatening power for protection or gain (Schweller, 1994); and hedging, which involves engaging multiple great powers simultaneously while avoiding firm commitment to any single actor (Goh, 2005; Haacke, 2019; Kuik, 2021). The hedging literature has advanced considerably since Medeiros (2005)

lamented that the concept was "highly underdeveloped in both international relations theory and security studies literature." Koga (2018) revisited the concept systematically, situating it within the balancing-bandwagoning spectrum and arguing that states adopt hedging as a third strategic option under conditions of power shift and strategic uncertainty. Crucially, Koga argues that economic and military indicators alone are insufficient to classify alignment behaviour, and that diplomatic factors must be incorporated as a complementary, if weaker, signal. Kuik (2021) (2021) developed this further, arguing that hedging is best understood not as a calculated grand strategy but as instinctive insurance-seeking behaviour driven by uncertainty about both the principal threat and the reliability of potential patrons, and that the concept travels beyond Southeast Asia to other regions facing analogous structural conditions. Despite this theoretical progress, systematic empirical measurement of alignment behaviour remains underdeveloped, particularly outside the East Asian context. Africa and Latin America, which are emerging as central arenas of geopolitical and geoeconomic competition, remain largely absent from the quantitative alignment literature [why do we refer to alignment literature here?]. The ALLIES project at the Royal Military Academy addresses precisely this gap, investigating how African and Latin American states navigate great power competition through multilayer network analysis and qualitative case studies. This PhD research is embedded within the ALLIES framework and contributes to its first research objective: classifying the alignment behaviour of weaker states toward the EU, Belgium, and the competing great powers. It does so by developing a novel Multilayer Alignment Index (MAI) for Latin America that addresses three persistent limitations in existing measurement approaches [no. focus on network analysis]. The first limitation concerns the conflation of structural dependency with voluntary political alignment. Existing approaches that use bilateral trade flows and arms procurement data as alignment proxies risk measuring market structure rather than strategic preference. China is the dominant trading partner for the vast majority of Latin American states regardless of their political orientation, a reflection of global commodity demand rather than deliberate alignment choice. US arms exports reflect supplier dominance and interoperability ecosystems rather than bilateral political deference. A country purchasing US F-16 fighter jets or exporting copper to Chinese refineries is not necessarily signalling political alignment in either direction. The second limitation concerns the conflation of reactive adaptation with strategic hedging. As Koga (2018) notes, identifying hedging requires distinguishing it from mere "waiting for balancing," and Kuik (2021) (2021) emphasises that states hedge because structural conditions compel them to, not because they have designed a deliberate multi-vector strategy. The empirical record from Latin America 2020-2026 illustrates this distinction sharply: Bolivia's rapid reorientation toward Washington following the 2025 election reflects reactive adaptation to domestic political change rather than a sustained hedging strategy, while Panama's simultaneous hosting of US military exercises and Chinese port concessions better approximates genuine strategic hedging. Collapsing these cases into a single alignment continuum obscures analytically important variation. The third limitation is the exclusive focus on the US-China dyad, which excludes other strategically significant actors. Russia's security presence in Venezuela, Cuba, and Nicaragua; the EU's normative and investment engagement through the Global Gateway; Turkey's expanding diplomatic and commercial network across the region; and Iran's influence operations through the Venezuelan diaspora all constitute alignment-relevant relationships that a strictly bilateral framework cannot capture.

Theoretical Contribution

The central theoretical contribution is a distinction between three forms of alignment behaviour that existing frameworks collapse into a single continuum. Structural dependency refers to inherited trade, financial, and security ties that constrain short-term choices regardless of political preference, such as Argentina's reliance on China's currency swap line or Ecuador's dependence on Chinese shrimp markets, despite both governments' pro-US orientations. Reactive adaptation describes alignment shifts driven by external actor behaviour rather than deliberate strategic design, as illustrated throughout the empirical record compiled in the accompanying political developments document: the DRC's drift toward China during Western disengagement under Kabila, Bolivia's rapid reorientation toward Washington under Paz following decades of Morales-era estrangement, and Honduras's alignment with Washington following Trump's explicit electoral intervention and the Hernández pardon. Strategic hedging, by contrast, denotes deliberate, sustained management of relationships with competing great powers to preserve policy autonomy, as illustrated by Panama's simultaneous hosting of US military exercises and Chinese port concessions, or Peru's purchase of US F-16s alongside its acceptance of the Chinese-financed Chancay mega-port.

This typology connects to Kuik's (2021) argument that hedging is instinctive rather than strategic, driven by uncertainty about both patron reliability and threat magnitude, and to the Latin American autonomy tradition described by Sims and Lee (2025) , which frames strategic ambiguity as a structural response to hierarchical pressure rather than a coherent foreign policy design. The empirical record from Latin America 2020-2026 provides important new evidence for this theoretical debate: in multiple cases, what appears as multi-vector engagement is better explained by the simultaneous pull of Chinese economic indispensability and US political coercion, a dynamic that intensified dramatically following the Trump 2025 National Security Strategy's explicit invocation of a Monroe Doctrine corollary.

Measurement Framework

The proposed Multilayer Alignment Index (MAI) replaces the Sims and Lee bilateral trade/arms dyad with a set of indicators selected specifically to capture voluntary political alignment rather than structural dependency. The indicators are grouped into four layers, each capturing a distinct dimension of alignment behaviour.

The first layer covers diplomatic and institutional signalling. This includes UNGA voting patterns on contested resolutions, specifically votes on Ukraine, Gaza, and human rights mechanisms, membership and participation in the Americas Counter-Cartel Coalition (ACCC), which currently comprises 19 member nations and serves as an explicit US alignment instrument, membership in the World AI Cooperation Organisation (WAICO), which Beijing launched in 2026 in explicit competition with the US-anchored Pax Silica framework, withdrawal from or continued membership in the International Criminal Court given US pressure to exit the institution, and free trade agreements with China, which Beijing has concluded with Chile, Costa Rica, Ecuador, Nicaragua, and Peru.

The second layer covers economic and financial alignment signals that go beyond raw trade volumes. This includes the stock of Chinese foreign direct investment relative to GDP, which captures structural economic penetration more meaningfully than bilateral trade flows, the existence of BRI membership and the status of specific BRI projects, Huawei's penetration of national 5G infrastructure, the imposition of US tariffs under Section 232 or equivalent measures as a proxy for Washington's assessment of each country's alignment, and receipt of US DRL (Bureau of Democracy, Human Rights and Labour) financing, which signals US political investment in a country's domestic politics.

The third layer covers security alignment signals. This includes membership in the ACCC and the Shield of the Americas, engagement with private security contractors linked to US strategic interests such as the Blackwater-affiliated arrangement in Ecuador, bilateral security agreements with the US, and the nature and conditionality of military assistance.

The fourth layer covers what this research terms coercion-response indicators, a novel category not present in existing frameworks. These capture cases where US financial or diplomatic pressure has produced measurable changes in alignment behaviour: currency interventions and IMF programme conditionality linked to political outcomes as documented in Argentina and Bolivia, visa revocations targeting officials whose decisions conflicted with US interests as documented in Peru, Panama, Chile, and Colombia, and the rescission of Chinese infrastructure contracts following US pressure as documented in Panama and Costa Rica.

Methodology

The index will be constructed for 20 Latin American states over the period 2020-2026, using a multilayer network approach implemented in R with the igraph and backbone packages, and visualised in Gephi. Each indicator layer constitutes a separate network, with edges weighted according to the strength of alignment signal. The multilayer structure allows the analysis to identify countries that appear aligned in one dimension but not others, which is precisely the analytical space where strategic hedging is most visible, and to distinguish countries whose multivector engagement reflects deliberate hedging from those whose apparent hedging is an artefact of structural constraints.

Community detection algorithms will be applied to each layer separately and to the aggregate network to assess whether distinct blocs emerge and whether they are stable across layers. Betweenness centrality will be used to identify structural brokers. The temporal dimension will be incorporated through a panel design that tracks changes in alignment scores across the six-year period, allowing the research to identify whether shifts are reactive to external events, such as electoral interference, tariff impositions, or Chinese infrastructure deals, or reflect more sustained strategic repositioning.

The empirical base will draw on primary sources including UNGA roll-call voting records, SIPRI, UN Comtrade, AEI China Global Investment Tracker, ACCC membership records, Boston University Global Development Policy Center Chinese lending data, and the primary source political developments analysis accompanying this proposal, which documents alignment shifts across all 20 countries in real time through October 2026.

Expected Contribution

This research makes three distinct contributions. Theoretically, it provides the first systematic operationalisation of the distinction between reactive adaptation and strategic hedging in Latin America, addressing what Sims and Lee acknowledged as the most important unresolved conceptual problem in their framework. Methodologically, it introduces a multilayer network approach to alignment measurement that is more sensitive to the multidimensional nature of great power competition than existing single-index approaches. Empirically, it produces a comprehensive, reproducible alignment dataset for Latin America 2020-2026 at a moment when the region is undergoing its most significant geopolitical reorientation since the Cold War, providing a baseline against which future studies can measure the trajectory of US-China competition in the hemisphere.

Comments

why do we focus on weaker states? is Brazil a weak state? if not, I still want to include it.

values: ICC membership/withdrawal

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